

Homework 10

Due: Tuesday 5/27 at noon

Complete the following and submit it to me using Google Classroom. Your submission must be in an electronic form that supports cut and paste (.txt, Word document, Google Doc, etc). DO NOT submit a PDF or image of your problem.

The problems you submit will be used as the basis of the in-class activity on Wednesday 5/28. If you don't want your problem distributed to the rest of the class (anonymously), then indicate this on your submission. The filename should include your name or user ID.

1. (10 points) Write a new design problem along the line of the ones we've been doing in class. It should include sufficient detail to have a specific solution using ADTs and data structures discussed in class (List, Set, Stack, Queue, Iterator, Map, Priority Queue).

Include a solution to your design problem. This should include the name of the data structure and how it can be used to implement each of the operations required in your problem description. Be sure to specify the types of things it holds (list of strings, map from string to integer, set of **Person** objects, etc). (Note that I'm NOT looking for actual code. Just tell me in English what data structure(s) to use and how it/they would be used to support the operations required by your design problem.)